Our Precious Planet

4th Claremont Cub Pack • Wednesday, 19 August 2026 • 17h30 – 19h00
Duty Six: Green Meeting Leader: Rob

Time	Activity and Venue	Leader	Equipment
17h15	**Beforehand** Flag pole, totem and Shere Khan skin set out. Attendance register. Set up the three bases (see checklist). Mix the bucket of "dirty" water. Put one egg shell and one stick of chalk into vinegar — these run all evening and get shown at report back. **Duty Six: Green**	All	Flag & pole, totem, tiger skin, register
17h30 – 17h42	**Opening and Grand Howl** Inspection: Claws and Paws. Birthdays: Micah Densmore **Theme introduction:** "We have one planet and no spare. Tonight we find out how we make it dirty, whether we can undo it, and what a Cub can actually do about it."	Akela	Flag, totem, tiger skin
17h42 – 17h55	**Steam Release: Pollution Patrol** Newspaper "litter" scattered across the hall. Half the Pack are Polluters, half are the Clean-Up Crew. 90 seconds — the Crew get litter into the bins, the Polluters take it out again. Swap sides and replay. Final round: everyone is the Clean-Up Crew. Time how fast the Pack clears the hall. **The point:** one Cub can make a mess faster than three can clean it up — but when everybody helps, it takes no time at all. **If it is pouring:** run Four Corners instead — walls labelled RECYCLE, COMPOST, LANDFILL, REUSE. Also the filler if a base runs short.	Hathi + all	± 60 crumpled newspaper balls, 4 boxes/bins, cones, bibs or bands, 4 wall labels
17h55 – 18h05	**Leaping Wolf Presentation — Josie** Pack sits in a horseshoe on the floor. Josie has the floor for her Leaping Wolf presentation, then takes questions from the Cubs. Leaders sit down with the Pack rather than hovering. Sixers are responsible for their Six listening — brief them at inspection. **Have two questions ready** in case the Pack goes quiet, and make them specific to what she has done rather than "did you enjoy it". Finish with a proper cheer. A leader notes the date in her record book on the night. **Running late?** Take the time out of the discussion, not out of Josie's ten minutes and not out of the bases.	Josie (Akela to chair)	Projector

18h05 – 18h15	**Discussion: What is pollution?** • Water pollution — what causes it, can we undo it? • Air pollution — and how plants help fight it • Acid rain — what it is and what it damages • What can one Cub in Claremont actually do? Write the Pack's answers on the flip chart and leave it up for the bases.	Rob	Flip chart / whiteboard, koki pens
18h15 – 18h55	**Bases — Sixes rotate, 10 minutes each** **Base 1: Build a Water Filter** — SW: show how you could undo pollution in water. **Base 2: Make a Water Scope** — SW: observe life underwater. **Base 3: Acid Rain Test** — GW: what acid rain is and what it damages. One activity per base, so nobody is rushed. Plants vs air pollution (GW) and the diorama (GW) are handled at report back and as take-home. Full instructions in the leader notes overleaf. **Timekeeper: Akala.** 12 minutes.	All	See equipment checklist
18h55 – 19h00	**Closing Grand Howl** Reflection and notices. Reminder: interest badge night on 26 August — bring dust catchers and dioramas back for sign-off.	Akela	Flag, totem

Leader Instructions — Activity Details

Key Learning Outcomes

Silver Wolf	Gold Wolf
• Find out three things that can cause water pollution and show how you could possibly undo pollution in water. • Make a poster to examine insects OR make a water scope to observe life underwater.	• Find out how plants help fight air pollution and run an experiment to show this. • Find out about acid rain and demonstrate how it affects our planet. • Make your own diorama to show different types of habitats or environments (forests, grasslands, deserts, oceans).

Prepare Before the Meeting

- Tea indicator: brew four strong cups of ordinary black tea (Five Roses, no milk), let it cool and keep it in a bottle. Strong tea goes visibly pale the moment acid hits it, which is the whole trick. Rooibos works too but the colour change is weaker.
- "Dirty water": a bucket of water with garden soil, torn leaves, a squirt of cooking oil, a little dishwashing liquid and some glitter or shredded plastic.
- Ask parents the week before for clean 2ℓ cold drink bottles — you need about 12, plus shoeboxes.
- Cut the bottles yourself beforehand with a craft knife and sand the cut edges. Cubs do not cut bottles.

Steam Release: Pollution Patrol

Scatter the paper balls evenly across the hall floor. Mark two "landfill" boxes at each end.

Round 1: Polluters vs Clean-Up Crew, 90 seconds. Litter in a box counts as cleaned; Polluters may only remove one piece at a time and must scatter it, not throw it at people.

Round 2: swap roles, 90 seconds.

Round 3: whole Pack cleans. Call the time out loud so they can hear the difference.

Debrief in one minute — do not lecture. Ask: which round was easiest? Why? Where does the litter in our street actually end up? (Stormwater drains → the Liesbeek → the sea.)

Wet weather alternative / spare-time filler: Four Corners

Same shape as the North-South-East-West game the Pack already knows. Label the four walls RECYCLE, COMPOST, LANDFILL and REUSE. Call out an item; Cubs run to the corner where it belongs. Last one there is out, or play with two lives.

Items to call: banana peel, cooldrink can, tea bag, broken mug, glass bottle, jam jar, chicken bones, newspaper, plastic straw, egg shells, old T-shirt, dead leaves, polystyrene tray, nappy, milk carton, cardboard box, cooking oil, dog waste, bread crusts, bottle caps.

Argue about the tricky ones: a jam jar could be recycle or reuse. Polystyrene and nappies are landfill and always will be. Cooking oil goes to neither — never down the drain.

Discussion — what is pollution?

Get three causes out of the Cubs before you tell them any. Prompt if needed:

- Litter and plastic thrown down, washed into stormwater drains and out to sea
- Sewage leaks and blocked drains
- Oil, petrol and brake dust washing off roads when it rains
- Factory waste pumped into rivers
- Soap, bleach and paint poured down the drain
- Fertiliser and pesticide running off gardens and farms

- Animal waste and dumped rubbish on the river banks

Base 1 — Build a Water Filter

Build the filter: take the top half of a cut plastic bottle and turn it upside down into the bottom half so it works as a funnel. Cubs pack it, in this order from the neck upwards — a plug of cotton wool or cloth, then fine sand, then coarse sand, then small stones. Pour a cup of the dirty water in slowly and let it drip through into a clear jar. Compare against a jar of the dirty water.

The lesson that matters: the filter took out the bits you can see. It did NOT take out the oil, the soap or the chemicals — the water is still not safe to drink, and it still smells. Some pollution can be cleaned up; most of it is far easier to never put there. Ask each Cub for one thing they will stop letting go down the drain.

Base 2 — Make a Water Scope

Take a bottle with the bottom already cut off. Stretch a square of cling film tightly over the cut end — smooth, no wrinkles — and hold it with a thick elastic band, then tape around it.

Push the film end gently into the basin of water and look down through the neck. The flat film cuts out the ripples so you can see straight down.

Stock the basin beforehand with stones, leaves, twigs and a few plastic bugs. Pond water and weed make it far better if you can scoop some on the way in, but tap water works and the base does not depend on it. A torch shone in from the side helps.

Ask: if we tipped the bucket of dirty water in here, what would you be able to see? What would happen to the creatures?

Base 3 — Acid Rain Test

Explain simply: burning coal, petrol and diesel sends sulphur and nitrogen gases up into the air. They dissolve in the water in the clouds, and the rain that falls is slightly acidic.

Test with the tea. Four clear cups, the same amount of tea in each, lined up against a sheet of white paper so the colour change shows:

- Tap water added — stays dark brown (neutral)
- Vinegar added — goes visibly pale, almost orange (acid)
- Lemon juice added — goes pale too, so the Cubs see it is not just the vinegar

Demonstrate the damage: the egg shell that has been sitting in vinegar since 17h15 is now soft, bubbling and pitted. That is what acid rain does to limestone buildings, statues and gravestones, and to the shells of snails and sea creatures. Put a fresh shell in a second cup so the Cubs see the bubbles start.

Also show: a leaf with a few drops of vinegar on it next to one with water on it. Acid rain damages leaves and makes the soil poor, which is why forests downwind of power stations die back.

Equipment Checklist

Steam release

- ± 60 crumpled newspaper balls, 4 boxes or bins, cones, team bands

Base 1 — water filter

- 6 cut 2ℓ bottles, cotton wool, fine sand, coarse sand, small stones, clear jars, bucket of dirty water, funnel, jug

Base 2 — water scope

- 6 cut 2ℓ bottles, cling film, thick elastic bands, insulation tape, 2 large basins, pond water and waterweed, torch

Base 3 — acid rain

- Flask of strong black tea, white vinegar, lemon juice, clear plastic cups, sheet of white paper, 3 egg shells, a stick of chalk, a few leaves, rainwater if available, eye droppers

Quiz — Our Precious Planet

Use at report back, or keep for the make-up week on 16 September.

1. Name three things that can cause water pollution.
2. Where does the litter in our street end up when it rains?
3. What are the four layers, from the bottom up, in a bottle filter?
4. Your filtered water looks clean. Why is it still not safe to drink?
5. What gas do plants take in, and what gas do they give out?
6. Other than making oxygen, name one other way trees clean the air.
7. What causes acid rain?
8. Name two things that acid rain damages.
9. What does a water scope let you do, and why does it work?
10. Name one thing you are going to change at home this week.

Answers

1. Litter and plastic, sewage leaks, oil off the roads, factory waste, soap and chemicals down the drain, fertiliser and pesticide run-off, animal waste.
2. Down the stormwater drain, into the river, and out to sea — it is not cleaned on the way.
3. Cotton wool or cloth, fine sand, coarse sand, small stones.
4. The filter only removes the bits you can see. Oil, soap, chemicals and germs go straight through.
5. Takes in carbon dioxide, gives out oxygen.
6. Their leaves trap dust, soot and exhaust particles. (Also: shade and cooling, holding soil together.)
7. Burning coal, petrol and diesel puts sulphur and nitrogen gases into the air; they dissolve in cloud water and fall as acidic rain.
8. Limestone buildings and statues, gravestones, shells of snails and sea creatures, tree leaves, soil, fish in lakes and rivers.
9. Look under the surface of the water. The flat, tight film gives you a smooth window instead of a rippled surface.
10. Any honest answer — hold them to it and ask next week.